المعهد الفنى الصحى

الكترونات ودوائر كهربية

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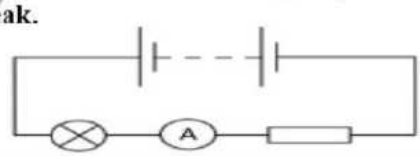
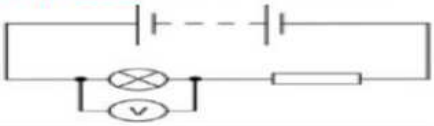
Multimeters

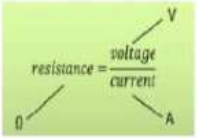
Definition	A multimeter measures several different electrical quantities, usually RMS voltage and RMS current.
Types	There are two kinds of multimeter analogue and digital. Both kinds of meter have a knob for selecting which quantity to measure range selection.
Sockets	Both meters have sockets for probes (Positive marked '+' and usually red), (negative marked '–' Common or COM and usually black). Some meters have a second positive socket which must be used for relatively high voltages.
When you measure current	The meter is part of the circuit as it is connected in series (operated as an ammeter). So, break the circuit at one point and connect the meter across the break. 
When you measure voltage	You are measuring the difference of electrical force at two points in the circuit. Measure the voltage across a component a voltmeter should be connected in parallel with the component. 
Test routine	1. Before you connect the meter to the circuit, set the selector knob to the range you want to use. 2. If the meter has auto-ranging, you need only select the quantity (current or voltage). 3. If you are not certain which range to choose, select the highest range. 4. If you need to change the range, disconnect the meter from the circuit first.

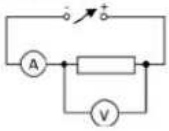
Difference between analogue and digital multimeter

Analogue Multimeter	Digital Multimeter
Using an analogue meter, touch the black (common) probe to the more negative point and red to the more positive. If you touch them the wrong way around, the needle swings below zero. Remove the probes immediately or you may damage the meter.	Digital meters often have auto-polarity. These work with the probes touching either way and display a minus sign to the left of the reading.
With an analogue meter, the reading is wrong if you do not look straight down at the scale. This is called a parallax error. To help you avoid a parallax error the meter has an arc of mirror on the scale.	Digital meters may be auto-ranging. They select the correct range automatically when a measurement is made.

Experiment 1. Use of meters, and Ohm's law

Determine the value of a resistance using voltage, and current measurements, and directly with an ohmmeter

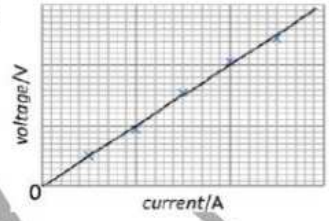




(a) Connecting the resistor to a DC variable voltage power supply,

and taking a range of measurements of voltage across and current in the resistor.

- The graph of voltage against current for this resistor is straight line and its gradient (slope) is equal to the resistance of the resistor.



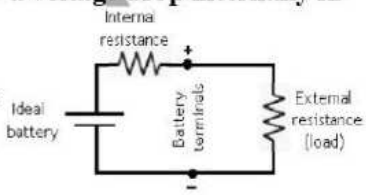
(b)Verify the found value in (a) using a multimeter operating as an ohmmeter.

Experiment 2. Current and internal resistance

(a) All power sources have an internal resistance.

(b) When current is flowing through the power source (e.g. a battery), there will be a voltage drop internally in the power source due to internal resistance. therefore,

(c) The voltage available is smaller than the unloaded voltage (or nominal voltage).



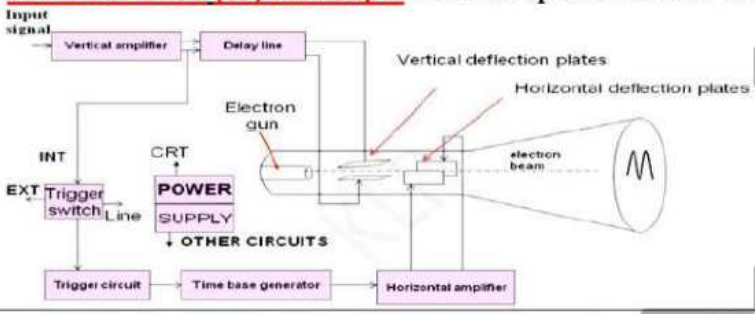
Experiment 3. Battery test

Battery capacity is measured in ampere-hours, meaning number of hours it will deliver 1 amp.

The capacity is also temperature dependent. The capacity of a battery can be measured by the maximum charge it can give.

Oscilloscopes

Definition	**An oscilloscope, also called Cathode Ray Tube (CRO), provide recorded visualization of voltage and current waveforms. A CRO allows the amplitude of displayed as a function of time; it gives visual representation of time varying signals.** ➢ **The oscilloscope is the basic instrument for the study of all types of waveforms.**
Uses	**The oscilloscope can be used to measure quantities such as, peak voltage, frequency, phase difference, pulse width, delay time, rise time and fall time.**
Applications of CROs	**1. Waveform analysis. 2. Amplitude measurement. 3. Time period measurement.** **4. Frequency measurement. 5. Phase angle 6. Current measurement.**
Main components	**1- Cathode Ray Tube (CRT):** This is the heart of CRO. CRT consists of electron gun, which emits electrons to strike the phosphor screen through deflecting plates. **2- Deflecting Plates:** are two types, horizontal deflecting plates moves the beam from left to right or right to left horizontally & vertical deflecting plates move the beam up and down. **3- Time Base Generator:** is used to generate the saw tooth voltage in the horizontal section. **4- Horizontal Amplifier:** is used to amplify the saw tooth voltage before it is applied to horizontal deflection plates. **5- Vertical Amplifier:** This is a wide band amplifier used to amplify signals in the vertical section.

	6- Power Supply: High voltage supply (HVS): it is applied to CRT of the order of 1000V to 1500V. & **Low voltage supply (LVS):** it is required for heater of electron gun. **7- Delay Line:** It used to delay the signal time in the vertical sections. **8- Trigger Circuit:** controls the instants at which the sweep voltage is to be applied to the horizontal deflecting plates. used to convert the incoming signal into trigger pulses, so that the input signal and the sweep frequency can be synchronized. **9- Fluorescent Screen:** is inside face of CRT and coated with a fluorescent material called phosphor. **10- Glass Tube (or) Envelope:** All the components of CRT are enclosed in an evacuated glass tube. 
The peak to peak value of voltage	$V_{p-p} = \left(\frac{volts}{div}\right) No.of\ vertical\ divisions$
Periodic time	The waveform must be displayed such that a complete cycle is displayed on the CRT screen. $T = \left(\frac{time}{div.}\right) \frac{No.of\ div.}{cycle}$
Frequency measurement	$f = \frac{1}{T}$

Function and use of controls and input terminals of CRO

1. Horizontal amplifier input (X-input): this is used for connecting the external horizontal signal as an input for horizontal amplifier.

2. Vertical amplifier input (Y-input): Signal under testing connected to the vertical amplifier input.

3. Basic controls: Intensity (brightness) – Focus (sharp trace on the screen) – Astigmatism (another focus control, a very sharp spot can be obtained both in the center and also at the edges of the screen.

4. X-shift or horizontal position control moving the pattern left to right and right to left on the screen.

5. Y- Shift or vertical position control moving the pattern up and down on the screen.

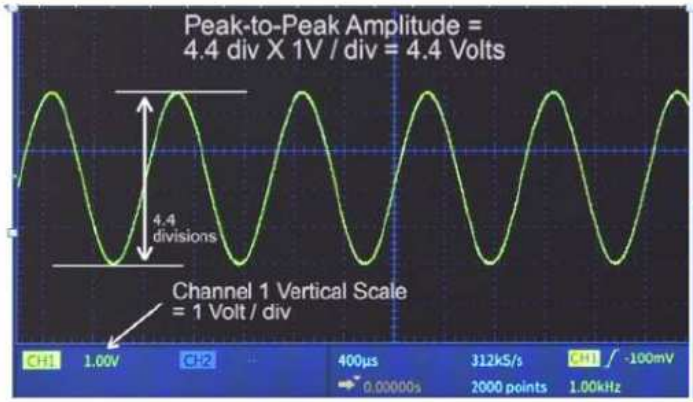
6. Time/Div. control: adjusts the horizontal gain that is the width of the pattern; time division.

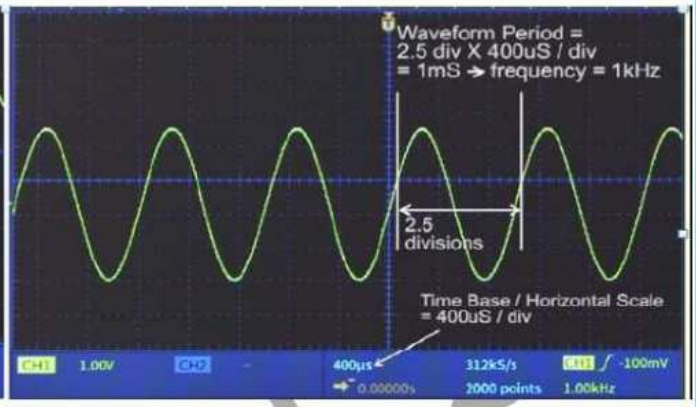
7. Volts/Div. control: This control adjusts the vertical gain that is the height of the pattern.

8. Variable speed: adjusts the internal sweep frequency of vertical input signal.

9. EXT position: the horizontal deflection system is fed from an external input.

10. Coupling (AC-DC-GND): It the mode of input coupling i.e., whether the input to be A.c coupled, D.c coupled or GND.





Example: Determine the pulse amplitude or peak (V_p), peak to peak (V_{p-p}), RMS value, Periodic time (T), frequency (f), phase difference, rise time (t_r) and fall time (t_f) of the waveform:

Characteristics of simple time-varying signals:

- Sources that vary as a function of time (called AC or alternating current sources).
- A Periodic time-varying signal is defined as the time within which the signal repeats.
- The frequency can be calculated from:

$$f(Hz) = \frac{1}{T(\sec)} \quad \text{,or} \quad \omega(radians/\sec) = \frac{2\pi}{T(\sec)}$$

The amplitude of varying signal (V_{peak}) is characterized in one of several ways:

- A first way: If we describe the signal as $V(t) = V_p \sin(\omega t)$
- A second way: to describe the signal is in terms of its peak-to-peak voltage, V_{P-P}.
- A third way, is based upon the ability of a source to deliver power to a resistor.
- Peak-To-Peak Voltage, V_{P-P} This is the voltage difference between maximum and minimum value.
- Average power delivered to a resistor by a DC source is: $p = I^2.R$
- average power delivered to a resistor by a periodic current, i(t), is: $p = \frac{1}{T}\int_0^T i^2(t)R\,dt$

We can define an *effective current*, I_{eff}, for the AC source as the equivalent DC current that would deliver the same power to the resistor. $I_{eff} = I_{rms} = \sqrt{\frac{1}{T}\int_0^T i^2(t)\cdot dt}$

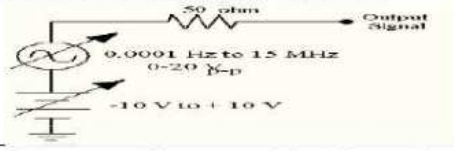
- The rms voltage, Vrms for a voltage signal of the form v(t) = Vm sin (ωt) as,

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T v^2(t)dt} = \sqrt{\frac{1}{T}\int_0^T V\sin^2(\omega t)dt} = 0.707V$$

-Example: rms voltage 115 VAC. This means that:

Vp = 115 / (0.707) = 162.6V and VPP = 2x162.6 = 325.2V

The signal generator

Definition	**The signal generator is a voltage source which can produce various time dependent signals waveforms from 0.0001 Hz to about 13 MHz.** 
Uses	**The signal generator can produce a variety of waveforms including sine, square, or triangle waves as the output signal. It can also produce a simulated cardiac signal for testing biomedical instrumentation.**

Instrument Transformers

Current transformers, (CT) together with voltage transformers (VT) (potential transformers (PT)), are known as instrument transformers.

	Current Transformers, (CT)	Potential Transformers (PT)
Definition & Uses	➢ **A current transformer (CT) is used for measurement of alternating electric currents.** ➢ **A current transformer (CT) is used when current in a circuit is too high to directly apply to measuring instruments.** ➢ **A current transformer also isolates the measuring instruments from what may be very high voltage in the monitored circuit.** ➢ **Current transformers are commonly used in metering and protective relays in the electrical power industry.** ➢ **A current transformer produces a reduced current proportional to the current in the circuit.**	➢ **Potential transformer or voltage transformer gets used in electrical power system for stepping down the system voltage to a safe value which can be fed to low ratings meters and relays.** ➢ **A voltage transformer theory or potential transformer theory is just like a theory of general-purpose step-down transformer.** ➢ **The two terminal voltages are in precise phase opposite to each other.** ➢ **PT has lower turns winding at its secondary. The secondary voltage of the PT is generally 110 V.** ➢ **Primary is connected across the phase and ground.** ➢ **The ratio of primary and secondary voltages of transformer is equal to the turns ratio.**

Current transformers, and Clamp meters: a current transformer has a primary winding, a magnetic core, and a secondary winding.

- **The alternating current flowing in the primary produces an alternating magnetic field in the core, which then induces an alternating current in the secondary winding circuit.**

The most common design of CT: Consists of a length of wire wrapped many times around a silicon steel ring passed 'around' the circuit being measured.

- **The CT's primary circuit therefore consists of a single 'turn' of conductor, with a secondary of many tens or hundreds of turns.**
- **For example, a 4000:5 CT would provide an output current of 5 amperes when the primary was passing 4000 amperes.**
- **Many analogue, and digital clamp meters, utilizes a current transformer for measuring AC current.**

Bridge Circuits

Definition	**Bridge Circuit is a null method, operates on the principle of comparison. That is a known (standard) value is adjusted until it is equal to the unknown value. Bridges are often used for the precision measurement of component values, like resistance, inductance, capacitance.**
General bridge circuit components	**The simplest form of a bridge circuit consists of a network of four resistance arms forming a closed-circuit A source of current is applied to two opposite junctions and a current detector is connected to other two junctions.**
Principle	**The bridge circuit operates on null detection principle and uses the principle of comparison measurement methods.**

Notes:

1. It compares the value of an unknown component with that of an accurately known standard component.

2. The accuracy of measurement depends on the bridge and not on the null detector.

3. When no current flows through the null detector, the bridge is said to be balanced.

4. The relationship between the component values of the four arms of the bridge at the balancing is called the balancing condition or the balancing equation.

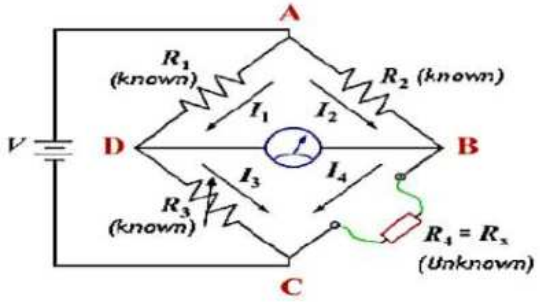
5. The balancing equation gives up the value of the unknown component.

Advantage of the bridges over other measuring instruments is that:

- **The balance condition (or equation) of any bridge circuit is independent of the source voltage magnitude and its impedance.**
- **The accuracy of the measurements is high in comparison with other methods of measurements.**

Classification of Bridge Circuit. Bridges are classified to two main categories; DC bridges, and AC bridges.

Wheatstone bridge: The Wheatstone Bridge is a DC type bridge usually used for measuring resistances.



Balance conditon: No potential difference across the galvanometer i.e. no current flow through it. That is called the null balance

Under this conditon: $V_{AD} = V_{AB} \Rightarrow I_1R_1 = I_2R_2$

Also: $V_{DC} = V_{BC} \Rightarrow I_3R_3 = I_4R_4$

Since, $I_1 = I_3$ and $I_2 = I_4 \Rightarrow$ Hence,

$$\frac{R_1}{R_3} = \frac{R_2}{R_4} \quad \text{or} \quad \boxed{R_x = R_4 = R_3\frac{R_2}{R_1}}$$

Limiting error of the known resistances:

$$R_x = (R_3 \pm \Delta R_3)\left(\frac{R_2 \pm \Delta R_2}{R_1 \pm \Delta R_1}\right)$$

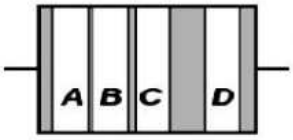
$$\boxed{R_x = R_3 \frac{R_2}{R_1}\left(1 \pm \frac{\Delta R_1}{R_1} \pm \frac{\Delta R_2}{R_2} \pm \frac{\Delta R_3}{R_3}\right)}$$

Example In the Wheatstone bridge circuit, R_3 is a decade resistance with a specified in accuracy ±0.2% and R_1 and R_2 = 500 Ω ± 0.1%. If the value of R_3 at the null position is 520.4 Ω, determine the possible minimum and maximum value of R_X

Example A Wheatstone bridge has a ratio arm of 1/100 (R_2/R_1). At first balance, R_3 is adjusted to 1000.3 Ω. The value of R_x is then changed by the temperature change, the new value of R_3 to achieve the balance condition again is 1002.1 Ω. Find the change of R_x due to the temperature change.

Color Coded Components

- Some components have color codes to indicate their mean value and its tolerance, or its rating.
- Color codes are used for resistors capacitors, inductors, diodes and others.



Resistor color coding

COLOR	FIRST BAND	SECOND BAND	MULTIPLIER	TOLERANCE
BLACK		0	$10^0 = 1$	
BROWN	1	1	$10^1 = 10$	
RED	2	2	$10^2 = 100$	
ORANGE	3	3	$10^3 = 1000$	
YELLOW	4	4	$10^4 = 10000$	
GREEN	5	5	$10^5 = 100000$	
BLUE	6	6	$10^6 = 1000000$	
VIOLET	7	7	$10^7 = 10000000$	
GREY	8	8	$10^8 = 100000000$	
WHITE	9	9	$10^9 = 1000000000$	
GOLD			$10^{-1} = 0.1$	5%
SILVER			$10^{-2} = 0.01$	10%
NO COLOR				20%

Color	Tolerance
Silver	± 10%
Gold	± 5%
Red	± 2%
Brown	± 1%
Green	± 0.5%
Blue	± 0.25%
Violet	± 0.1%
Gray	± 0.05%

Resistor Values

$$R = X \pm \Delta x\ \%$$

X: Nominal value, Δx: Tolerance

Test Questions

1. The value of brown, red, green, silver resistance is equal ---------- دور اول 2020

2. The value of resistance with color code blue, red, red, gold is equal -----دور اول2023 , دور ثان 2020

3. The color code of resistance 1.2 KΩ, 10% are --------,---------,----------,--------دور ثان 2020

4. The value of resistance with color code red, black, orange, red, gold = ----------

5. Determine the nominal resistance values of these resistors, given the band colors, and express the allowable tolerance in ohms. For example, a 25 KΩ resistor with a 10% tolerance rating would have an allowable tolerance of ± 2.5 KΩ. دور أول 2021

Red, Org, Blu, Gld =	Wht, Blu, Blk, Sil =
Brn, Blk, Grn, Sil =	Gry, Grn, Org, Gld =
Blu, Blk, Brn, Gld =	Org, Org, Gld =
Yel, Vio, Red, Sil =	Vio, Red, Sil, Gld =

6. The color code of resistance 4.75 KΩ, 10% =----------

7. What is the color code of 333kΩ,± 5% five band resistor?

8. In a 6 band color code the, what does the 6th band represent?(Temperature Coefficient – Multiplier – Tolerance - None of these)

9. What is the color code for a 5KΩ, tolerance ± 5% resistor?

10. What is the tolerance of a 3 band resistor? (±20%, ±0.1%, ±0.5%, ±0.25%)

11. What is the multiplier value for the color blue in multiplier band? (10K – 10M – 100K – 1M)

12. What is the value of a resistor which has first three color bands are Red, black, and Orange? (2KΩ - 20KΩ - 203Ω - 200KΩ)

13. Complete the following table: دور أول 2022

BAND 1	BAND 2	BAND 3	BAND 4	Value
Black	Brown	Brown	Gold	----------
Orange	White	Brown	Brown	----------
----------	----------	----------	----------	680k, 5%
----------	----------	----------	----------	1.2k, 10%
----------	----------	----------	----------	220. 20%

14. Complete the following table:

1st Digit Band 1	2nd Digit Band 2	Multiplier Band 3	Tolerance Band 4	Resistor Value
Black	Brown	Black	Gold	
Orange	White	Brown	Brown	
Brown	Black	Red	Gold	
Yellow	Violet	Green	Red	
Orange	Orange	Orange	Silver	
Brown	Black	Gold	Gold	
Brown	Black	Yellow	Silver	

1st Digit Band 1	2nd Digit Band 2	Multiplier Band 3	Tolerance Band 4	Resistor Value
				680kΩ, 5%
				1.2kΩ, 10%
				1.2Ω, 10%
				820Ω, 1%
				1.5MΩ, 10%
				1Ω, 5%
				3.9kΩ, 20%

Nominal Value	± Tolerance	Minimum value	Maximum value
680kΩ	5%	?	?
820Ω	1%	?	?
3.9kΩ	?	3.822kΩ	?

Sheet No. 2

Q 1: Complete these sentences:

1. Examples of insulators are ....., ..., ..., and .....
2. If enough energy is applied to the atom, some of the valence electrons will leave the atom. These electrons are called .....
3. A projection lamp runs on a 36 V supply and takes 10 A. So its power is ..... w.
4. ..... is a theoretical concept of an electric current or voltage supply (such as a battery) that has no losses.
5. ..... are materials with electrons that are loosely bound to their atoms, such as copper, silver, and gold.
6. An electric hair-clipper runs on 200 V, and is rated at 10 W. Its current =.....microamp.
7. If $\varphi = 0$ then the current and voltage are .....
8. Root Mean Square Value = $\sqrt{\frac{1}{T}\ldots\ldots\ldots\ldots\ldots\ldots}$
9. The relationship between frequency and periodic time is.....
10. Voltage measured in .....unit, and measured by ..... While Current measured in .....unit, and measured by.....

Q2. Put (✓) on true sentence and (X) on false sentence:-

1. Potential transformer is step-down transformer.
2. Oscilloscopes are used for measuring frequency.
3. A bridge circuit are used to determine the value of unknown components.
4. Potential transformers the two terminal voltages are in precise phase opposite to each other.
5. Current transformer is used to measure very low current.
6. The value of (orang – orang – brown – silver) resistance is 4.4 K ohms.
7. A multimeter can be used to measure frequency.
8. A current transformer use to measure DC current.
9. Potential transformer called voltage transformer.
10. Semiconductors are materials with electrons that are loosely bound to their atoms.

Q3. Choose the correct answer:

1. If you are not certain which range to choose, select the (highest – lowest) range.
2. The average voltage (equal – not equal – greater than) root mean square voltage.
3. Current transformer is step (up – down – matching) transformer.
4. In series circuit, there is .....for current (multiple – two path – one path)
5. Ohm's law, the current is (equal – inversely – directly) proportional to the applied voltage.
6. One of the applications for CRO are.....(frequency – impedance – capacitance)
7. (DMM – AVO – CRO) recorded visualization of voltage and current waveforms.
8. The colors of 2200 ohm resistance is (red – green – blue)

Chapter 3 Wiring Components and DC circuits on Circuit Boards

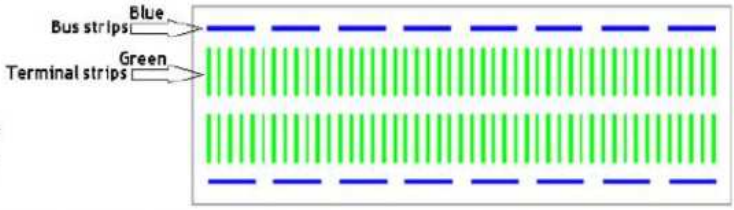
Types of Circuit Boards	Breadboard – Stripboard - Printed Circuit Board (PCB) - IC spacing PC board - Schmartboard - Perma-Proto boards - Breakout Boards – Arduino.

No.	Circuit Boards	Definition
1	Breadboard	Breadboard is a temporary connection between elements can be made without the need of any soldering for testing purposes or to try out an idea.
2	Stripboard	Unlike the breadboard, permanent soldered circuits can be constructed on stripboards it has parallel strips of copper track on one side. The strips are 0.1" (2.54mm) apart and there are holes every 0.1" (2.54mm).
3	Printed Circuit Board (PCB)	PCBs are multi-purpose circuit boards. With PCBs, permanent soldered circuit can be constructed. but their scale can be large compared with those constructed on stripboards. ➢ PCBs have copper tracks connecting the holes where the components are placed. They are designed especially for each circuit.
4	IC spacing PC board	These boards are very similar to the multi-purpose PCB boards, but these boards that have holes layouts that exactly fit certain integrated circuits.
5	Schmartboard (Radio Shack)	Schmartboards are special kind of "IC spacing PC Board", designed to make it easy to solder surface mount IC's in place.
6	Perma-Proto boards (Adafruit)	Another special kind of "multi-purpose/universal component PC board". This style of board exactly mimics تماثل تماما the layout "solderless breadboards" Their purpose is to allow a person who has built a circuit on a solderless breadboard to move all components and wiring directly from the solderless breadboard to the Perma-Proto board, soldering everything in place.
7	Breakout Boards	These are form of the "IC spacing PC board". They are designed to add header pin "legs" to a specific integrated circuit.
8	Arduino-type shields	These boards are designed with all needed headers to plug right into (on top of) a particular microcontroller such as the Arduino UNO, Arduino ATMega2560.

Breadboards are used for testing and experimenting with electronic circuits, and construction of educational projects.

Breadboards advantages: 1. No soldering is required. **2.** All the components can be re-used afterwards without the need of de-soldering. **3.** Easy to change connections and replace components.

- **Any breadboard has** many strips of metal which run underneath the board that connects the component.
- Each strip is a connection. So, the components connected to a certain strip are connected to each other.



1- The blue strips used for connecting the batteries. These strips are called **bus strips.**

2- The green strips are for the components are called **terminal strips.**

3- Integrated circuits are placed with one row of pins on each side of the division in the middle, because each side of the chip has pins this prevents them from becoming connected in series.

Connection of Components

- **An electric circuit is a complete path from the positive terminal to the negative terminal of a power source.**
- **Electrical components can be simply connected in either series or parallel.**
- **In series circuits there is only one path for current in a series circuit, so the current is the same at all points, while the sum of voltages across all the series connected components equals to the supply voltage.**
- **In parallel circuit the sum of currents in all parallel branches equals to the supply current, while the voltage across each branch equals to the supply voltage.**

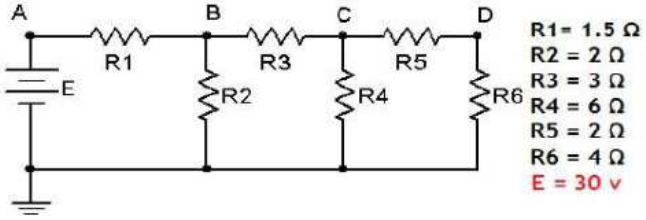
DC resistive circuits

Quiz

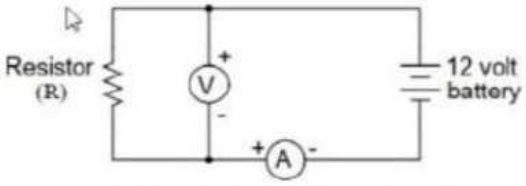
1. If voltages are identical, then how are the components connected?

2. If currents are identical, then how are the components connected?

3. Calculate the equivalent resistance, total current, total power dissipated.

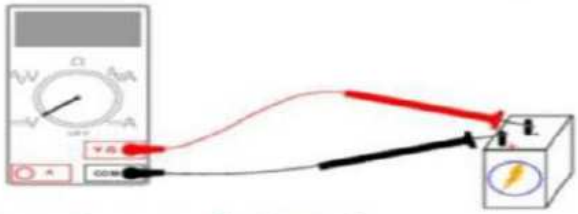


4. What is the value of this resistor in ohm (2700 – 27000 – 27 – 270)

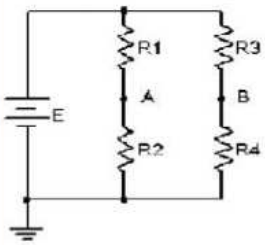


Voltmeter indication = 12.3 volts
Ammeter indication = 4.556 milliamps

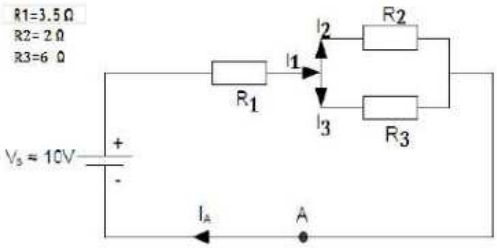
5. Reading of this device is (-6 , 3 , 6 , 9)

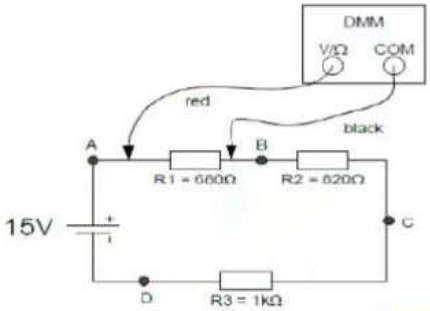


6. Calculate the equivalent resistance, total current, total power dissipated.

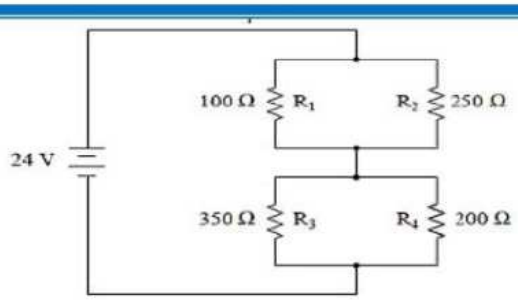


R1=2 Ω
R2=4 Ω
R3=1 Ω
R4=2 V
E= 24 v

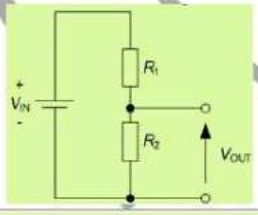
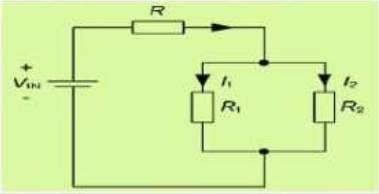




7. In the circuit shown, calculate the current flowing in the circuit and the power.



DC voltage divider circuit & DC current divider circuit

	Definition	Equation	Circuit
DC voltage divider	**Voltage Divider circuits are used to produce different voltage levels from a common voltage source but the current is the same for all components in a series circuit.**	$\frac{Vout}{Vin} = \frac{R2}{R1 + R2}$ $V_{OUT} = \frac{R_2}{R_1 + R_2} V_{IN}$	
DC current divider	**Current Divider circuits have two or more parallel branches for currents to flow through but the voltage is the same for all components in the parallel circuit.**	$I_1 = I \frac{R_2}{R_1 + R_2}$ $I_2 = I \frac{R_1}{R_1 + R_2}$	

1. How much current will flow through a 20Ω resistor connected in series with a 40Ω resistor when the supply voltage across the series combination is 12 volts dc? Also calculate the voltage drop produced across each resistor. Using the voltage division rule.

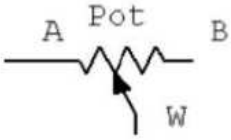
3. A 20Ω resistor is connected in parallel with a 60Ω resistor. If the combination is connected across a 30v battery supply, find the current flowing through each resistor and the total current supplied by the source. Answ. 1.5A, 0.5A, 2A

Potentiometers and Rheostats.

Definition	**A potentiometer is a variable resistor, with three terminals. It has a small adjustable screw that when turned, can adjust the value of the resistance from zero up to the nominal value.**

A potentiometer is **a three terminal** resistive device. **The outer terminals** present a constant resistance which is the nominal value of the device. **A third terminal**, called **the wiper** arm, is a contact point that can be moved along the resistance. These three terminal configurations are used typically to **adjust voltage via the voltage divider rule**.

- If only a single outer terminal and the wiper are used, is referred to as **a rheostat**. These may be placed in-line with a load to control the load current, **the greater the resistance, the smaller the current**.

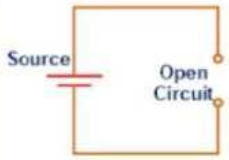
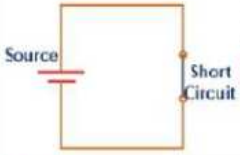


R(A to B) = the nominal value of resistance
R(A to W) + R(W to B) = nominal resistance

Difference between Open and Short Circuit

An open circuit implies that the two terminals are points are externally **disconnected.**

A short circuit implies that the two terminals are externally **connected.**

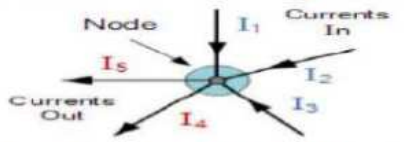
Open Circuit	Short Circuit
 Ω for open → Infinite Current for open → I = V/R = V/∞ = 0 Vo.c = Vs	 Ω for short → Zero Current for short → I = V/R = V/0 = ∞ Vs.c = Zero
Resistance: The resistance between two terminals of open circuits is **infinite.**	**Resistance:** The resistance between two terminals of shorts circuits is ideally **zero.**
Current: open circuit means that **zero current.**	**Current:** short circuit means, a very high amount **(infinite)** of current passing through the circuit.
Voltage: The voltage across terminals of the open circuit is **equal to supply voltage.**	**Voltage:** The voltage across short circuit terminals is **zero.**

Chapter 4 Electrical DC Circuits

Ladder networks are comprised of a series of alternating series and parallel connections.

Bridge networks typically make use of four elements arranged in dual series and parallel configuration.

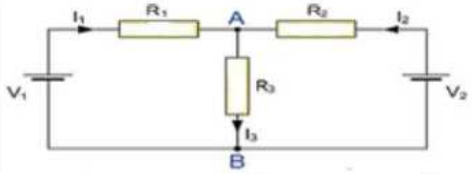
Kirchhoff's Circuit Law:

Kirchhoff's First Law – The Current Law, (KCL)	States that the "total current or charge entering a junction or node is exactly equal to the charge leaving the node". This means that in every moment of time the total current/charge flow through the node is 0.	**Kirchhoff's Current Law equation**  $I_1 + I_2 + I_3 + (-I_4 + -I_5) = 0$
Kirchhoff's Second Law – The Voltage Law, (KVL)	States that "in any closed loop network, the total voltage around the loop is equal to the sum of all the voltage drops within the same loop" which is also equal to zero.	**Kirchhoff's Voltage Law equation** Σ EMF= ΣV.D ΣEMF=ΣI.R

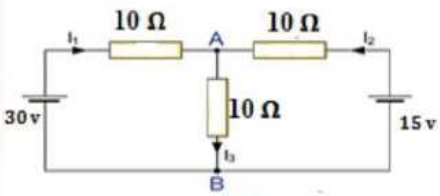
In other words, the algebraic sum of ALL the currents entering and leaving a node must be equal to zero, this idea by Kirchhoff is commonly known as the "Conservation of Charge".

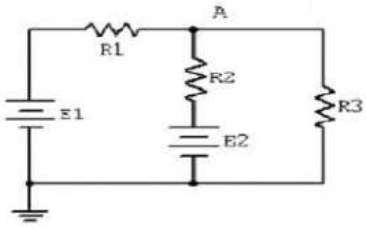
In other words, the algebraic sum of all voltages within the loop must be equal to zero. This idea by Kirchhoff is known as the Conservation of Energy.

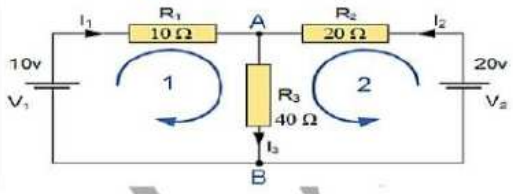
Example No1: Write the Kirchhoff's voltage and Kirchhoff's current equations of the following circuit.



Example No2: Find the current in each branch.







Capacitors and Inductors

- The equivalent of a series connection of inductors is the sum of the values.

- For a parallel connection of inductors either the product-sum rule or the "reciprocal of the sum.

- The equivalent of a parallel of capacitors is the sum of the capacitances.

- For series connection must be treated with the product-sum or reciprocal rules.

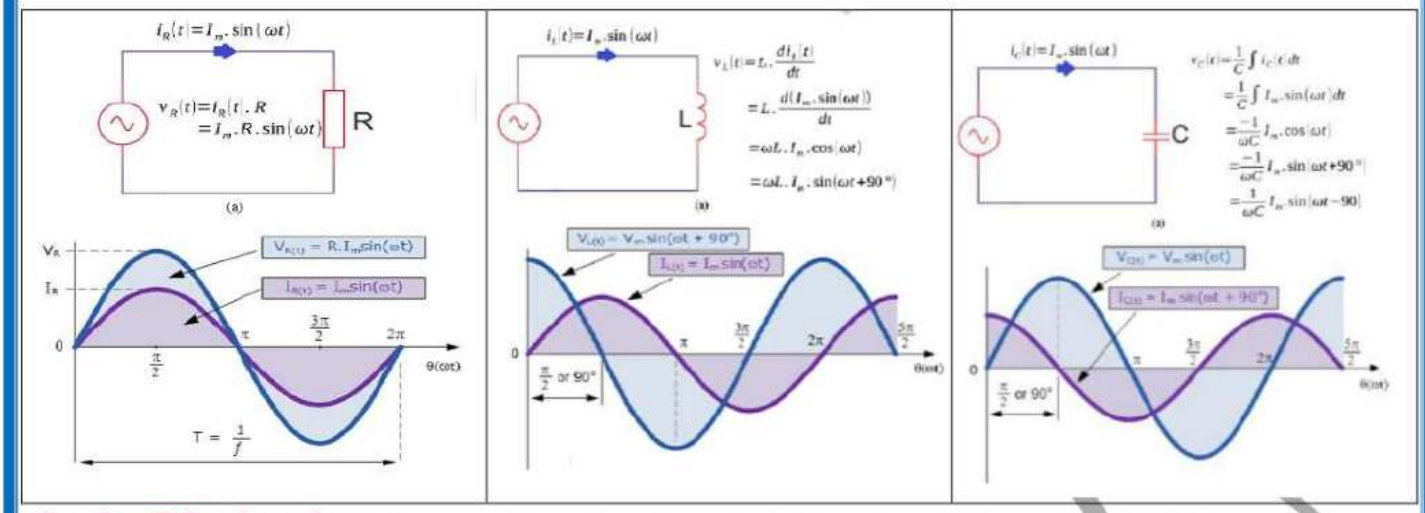
- In the steady state, inductors may be treated as shorts (as a small resistance known as the coil resistance, R-coil)

- In the steady state capacitors may be treated as opens.

Chapter 5 AC Electrical Circuits

The passive elements: *(i.e. resistance, inductance, and capacitance).*

Resistors	Inductors	Capacitors
The voltage across the resistance, and the current flow through it are in-phase.	The current lags the voltage by 90°	The current leads the voltage by 90°
The phase difference is zero.	The phase difference is 90° lag.	The phase difference is 90° lead.
	The "resistance" of the inductor in sinusoidal steady state is ωL; this value is called the inductive reactance (X_L) of the inductor L.	The "resistance" of the capacitor in sinusoidal steady state is $1/(\omega C)$; this value is called the capacitive reactance (X_C) of the capacitor C.
	Inductance is a parameter as it is based on the coil design, while the inductive reactance is a variable as it is based on the external frequency applied on the inductor.	Capacitance is a parameter as it is based on the capacitor design, while the capacitive reactance is a variable as it is based on the external frequency applied on the capacitor.



Series RL circuits: The current lags the voltage, but with a phase difference $0 < \phi < 90^o$

The exact value of the phase difference ϕ is obtained as: $tan\emptyset = \frac{VL}{V_R} = \frac{X_L}{R}$

Series RC circuits: The current leads the voltage, but with a phase difference $-90^o < \phi < 0^o$

The exact value of the phase difference ϕ is obtained as: $tan\emptyset = \frac{Vc}{V_R} = \frac{-X_c}{R}$

Series RLC circuits:

1- The phase difference depends on the relative value of Xc and X_L.

2- If the capacitive reactance is higher that the inductive reactance, then the circuit behaves as a capacitive circuit where the current leads the voltage by an angle ϕ.

3- If the inductive reactance is higher that the capacitive reactance, then the circuit behaves as a inductive circuit where the current lags the voltage by an angle ϕ.

3- In the case where both reactances are equal, then the circuit behaves as a resistive circuit where the current and voltage are in-phase i.e. zero phase difference.

The Resonance: In the case where both reactances are equal where the reactances cancel each other leaving the circuit with a minimum impedance equals to its ohmic resistance.

1. Resonance occurs in a series circuit when the supply frequency causes the voltages across L and C to be equal and opposite in phase.

2. At resonance in series RLC circuit, both inductive and capacitive reactance becomes equal in value, therefor both them cancel each other.

3. The point at which this occurs is called the Resonant Frequency point, (f_r) of the circuit.

Solve:

1: (a) Calculate the reactance of a coil of inductance 0.32 H when it is connected to a 50 Hz supply. (b) A coil has a reactance of 124 Ω in a circuit with a supply of frequency 5 kHz. Determine the inductance of the coil.

2: Determine the capacitive reactance of a capacitor of 10 μF when connected to a circuit of frequency (a) 50 Hz (b) 20 kHz.

3: A capacitor has a reactance of 40 Ω when operated on a 50 Hz supply. Determine the value of its capacitance.

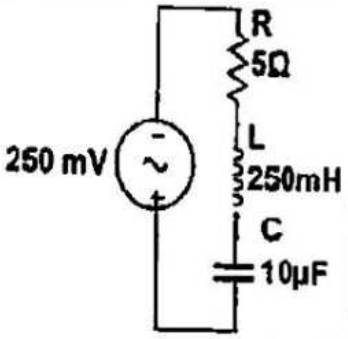
4: A coil has a resistance of 4 Ω and an inductance of 9.55 mH. Calculate (a) the reactance, (b) the impedance, and (c) the current taken from a 240V, 50 Hz supply. Determine also the phase angle between the supply voltage and current.

5: A resistor of 25 Ω is connected in series with a capacitor of 45 μF. Calculate (a) the impedance, and (b) the current taken from a 240V, 50 Hz supply. Find also the phase angle between the supply voltage and the current.

6: A series RLC circuit containing a resistance of 12Ω, an inductance of 0.15H and a capacitor of 100uF are connected in series across a 100V, 50Hz supply. Calculate the total circuit impedance, the complex impedance (Rectangular form, polar form and Exponential or Euler's form), the circuits current, power factor and draw the voltage phasor diagram.

7. Estimate the complex impedance of a RLC circuit ($R = 1\ k\Omega, L = 0.1\ H\ , C = 10\ \mu F$) which is powered by a sinusoidal source with frequency $f = 50\ Hz$.

8. Calculate all voltage drops and current in this RLC circuit at each of the given frequencies:



Frequency (HZ)	V_L (v)	V_C (v)	I_{total} (mA)
50			
60			
90			
100			
110			

Also calculate the resonant frequency of this circuit.

9: A coil of resistance 5 Ω and inductance 120 mH in series with a 100 μF capacitor, is connected to a 300V, 50 Hz supply. Calculate (a) the complex impedance, (b) the current flowing, (c) the phase difference between the supply voltage and current, (d) the voltage across the coil and (d) the voltage across the capacitor.

10. A series resonance network consisting of a resistor of 30Ω, a capacitor of 2uF and an inductor of 20mH is connected across a sinusoidal supply voltage which has a constant output of 9 volts at all frequencies. Calculate, the resonant frequency, the current at resonance, the voltage across the inductor and capacitor at resonance.